

Path-dependent Discrete Amortized Inference

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KEYWORDS — GFlowNets, State aliasing, Recurrent policies

TL;DR

- **Markovian** amortized samplers (e.g., GFlowNets) suffer from **state aliasing**: distinct states look (nearly) the same to the policy but require **sharply different** action distributions.
- we prove that Markovian forward policies are **provably less expressive** than their **path-dependent** counterparts,
- we **lift** the state graph by attaching a **learned latent variable** W_t to each state, keeping the joint process Markovian so that **all standard training objectives remain valid**,
- W_t evolves according to a learned dynamical system, which is parameterized by the newly introduced **Rotationary Self-Referential Weight Matrix** (R-SRWM),
- across grids, set generation, and sequence design, path-dependence yields **faster, stabler convergence** and an **improved fit** to the target distribution.

I. BACKGROUND: AMORTIZED SAMPLING WITH GFLOWNETS

Many core problems reduce to **drawing samples** from a distribution known only up to a constant,

$$\pi(x) = \frac{1}{Z} R(x), \quad Z = \sum_x R(x) \quad \text{intractable}, \quad (1)$$

where $\mathcal{X}$ is a **compositional** space: each x is built one component at a time from a seed s_o along a **trajectory** $\tau: s_o \rightarrow s_1 \rightarrow \dots \rightarrow s_T \rightarrow x$.

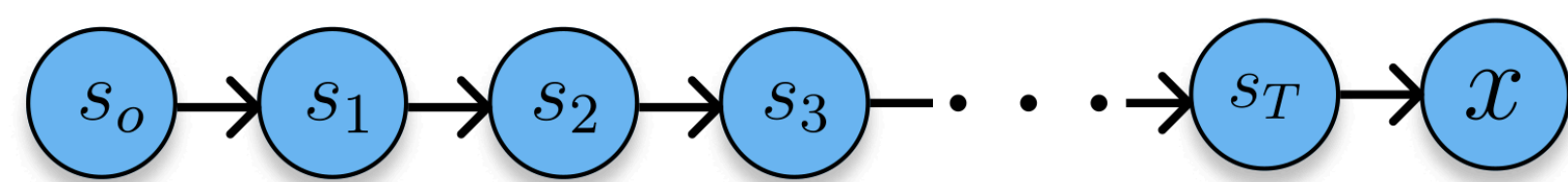


Figure 1: A **forward policy** $p_F(s_t | s_{t-1})$ navigates the state graph, adding one component per step. A **forward policy** $p_F(s_t | s_{t-1})$ rolls out from s_o to build x with marginal $p_F^T(x) = \sum_{\tau: s_o \rightarrow x} \prod_t p_F(s_t | s_{t-1})$. Our goal is to learn p_F so that $p_F^T(x) \propto R(x)$.

Because p_F^T is intractable, we introduce a **backward policy** $p_B(s_{t-1} | s_t)$ over the **parents** of a state and tie p_F to the reward through the **trajectory balance (TB)** condition,

$$Z \prod_t p_F(s_t | s_{t-1}) = R(x) \prod_t p_B(s_{t-1} | s_t). \quad (2)$$

Enforcing TB $\Rightarrow p_F^T(x) \propto R(x)$. We learn p_F, p_B, Z_θ by minimizing the expected **log-squared violation**,

$$\mathcal{L}(\tau; p_F, p_B) = \left(\log \left[Z_\theta \prod_t p_F(s_t | s_{t-1}) \right] - \log \left[R(x) \prod_t p_B(s_{t-1} | s_t) \right] \right)^2, \quad (3)$$

under an off-policy p_B (commonly ε -greedy, $(1 - \varepsilon)p_F + \varepsilon p_U$), i.e.,

$$p_F^*, p_B^* = \underset{p_F, p_B}{\operatorname{argmin}} \mathbb{E}_{\tau \sim p_B} [\mathcal{L}(\tau; p_F, p_B)]. \quad (4)$$

II. THE PROBLEM: STATE ALIASING

(Near) state aliasing. Different states look (nearly) the **same** to the policy, but require **different** action distributions. Equivalently, from the agent’s view the environment is **(almost) partially observable**—a plain network struggles to separate neighboring states when the optimal policy changes **sharply** between them (i.e., large Lipschitz constant).

A **Markovian** policy only sees the current state s_t . On the **LINES** environment (jump forward up to M steps or stop), we use the linear policy

$$p_F(p_{i+k} | p_i) = \operatorname{softmax}_k \left(\underset{\text{logits}}{W_i \psi(p_i)} \odot \underset{\text{mask}}{m(p_i)} \right), \quad (5)$$

with position embedding $\psi(p_i)$ and mask $m(p_i) \in \{1, -\infty\}^M$. It is **Markovian** when $W_i = W$ (same weights every step), and **path-dependent** when the weights **evolve along the trajectory**,

$$W_t = b a_t^\top, \quad a_t = a_{t-1} + (w_a^\top \psi(s_t)) v_a, \quad (6)$$

with b fixed and a_o, w_a, v_a learned.

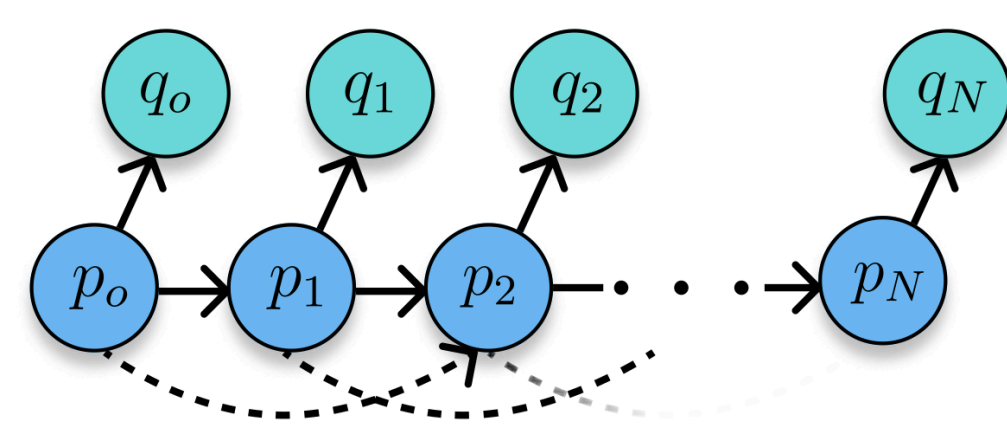


Figure 2: The **LINES** environment: from position p_i the policy either **jumps forward** (up to M steps) or **stops** at the terminal q_i .

Proposition 3.1 (expressivity gap). Let $\mathcal{R}_M$ and $\mathcal{R}_{PD}$ be the sets of distributions realizable by Markovian and path-dependent parameterizations. Then $\operatorname{span} \mathcal{R}_M$ is a **strict subspace** of $\operatorname{span} \mathcal{R}_{PD}$.

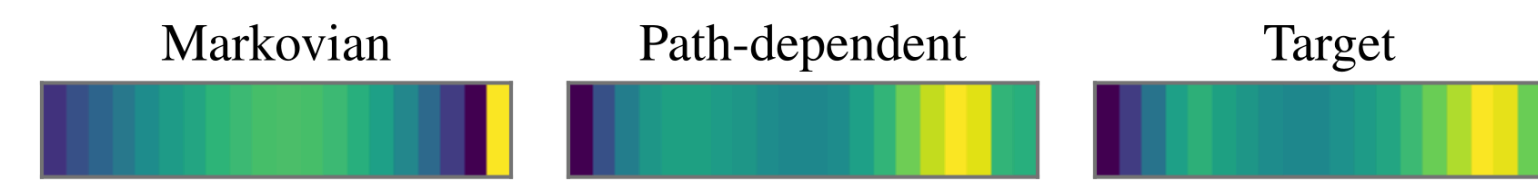


Figure 3: Target $R(q_i) = \sum_j w_j e^{-|i-k_j|}$: **Markovian** collapses to one mode; **path-dependent** fits the target.

III. STATE ALIASING IN PRACTICE

For a **sparse** target on **LINES** ($R(q_2) = R(q_{20}) = 1000 \cdot R(q_1)$), the optimal actions at p_2 and p_3 must differ **sharply**, even though the states are nearly indistinguishable.

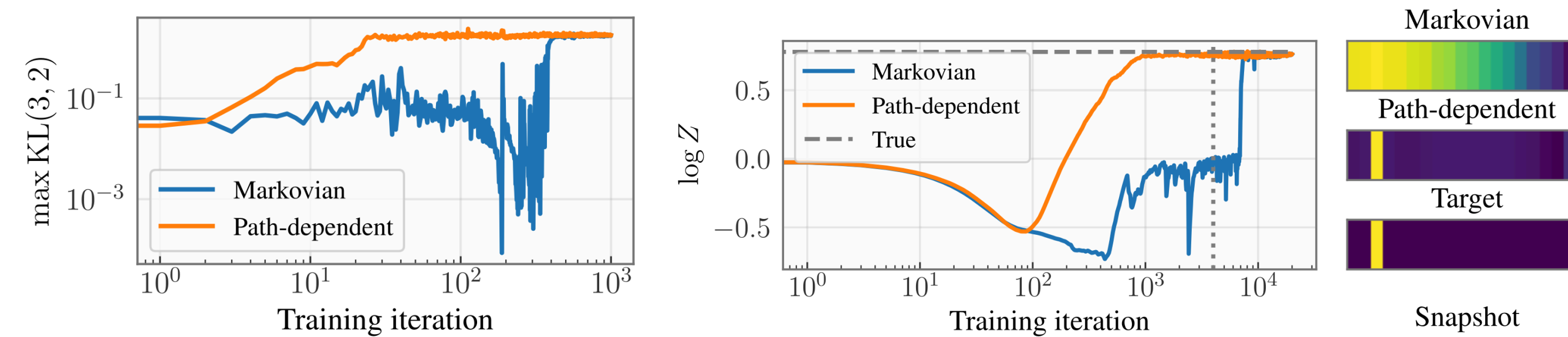


Figure 4: Separation of the learned policies at p_2 vs. p_3 (left) and convergence / goodness-of-fit (right): **Markovian** needs $\sim 100 \times$ more steps to tell p_2 from p_3 —slow and unstable.

On graph-building tasks a state is a graph G and a move adds one edge, scored by a message-passing network $p_F(G^{+(m,n)} | G) \propto e^{\zeta(f(x_m; G) \oplus f(x_n; G))}$. Message passing aggregates a **multiset** of neighbors, so its power is capped by the **1-WL** color-refinement test: some **non-isomorphic** graphs receive **identical** embeddings, forcing the policy to be **uniform** over states it should distinguish.

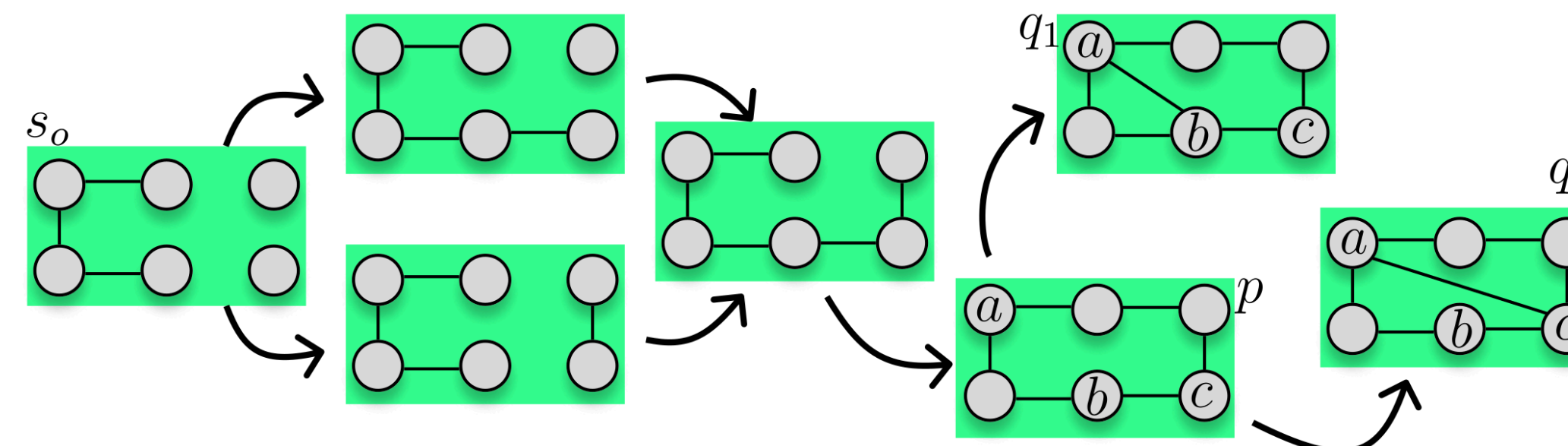


Figure 5: Building a graph edge-by-edge. At p , the pairs (a, b) and (a, c) look **identical** to a message-passing policy, forcing it **uniform** over q_1, q_2 ; knowing **when** each edge appeared breaks the tie.

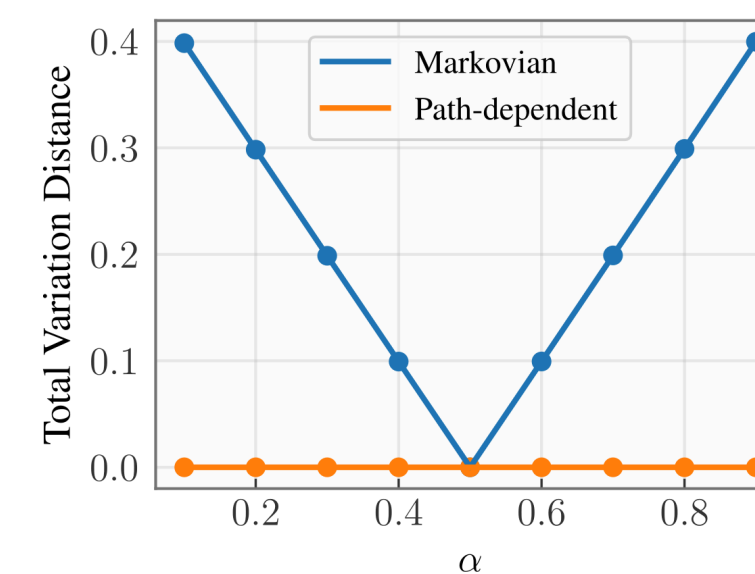


Figure 6: Target $R(q_1) = \alpha$, $R(q_2) = 1 - \alpha$: **Markovian** is stuck at uniform (error $|\alpha - \frac{1}{2}|$); **path-dependent** ≈ 0 .

Proposition 3.2. Path-dependence **strictly boosts expressivity** in graph-structured tasks—knowing **when** each edge appeared breaks ties that a Markovian graph network cannot.

IV. PATH-DEPENDENT AMORTIZED SAMPLING

We **lift** the state graph by attaching a latent **memory** $W_t \in \Omega$ to every state, updated recurrently as the object is built,

$$W_t = W_{t-1} + \varphi(s_t, W_{t-1}), \quad (7)$$

and feed it to the action head: $p_F(\cdot | s_t) \rightarrow p_F(\cdot | s_t, W_t)$. This mirrors **Hamiltonian Monte Carlo**—augmenting the state with a momentum-like variable that carries memory—but for **discrete** spaces.

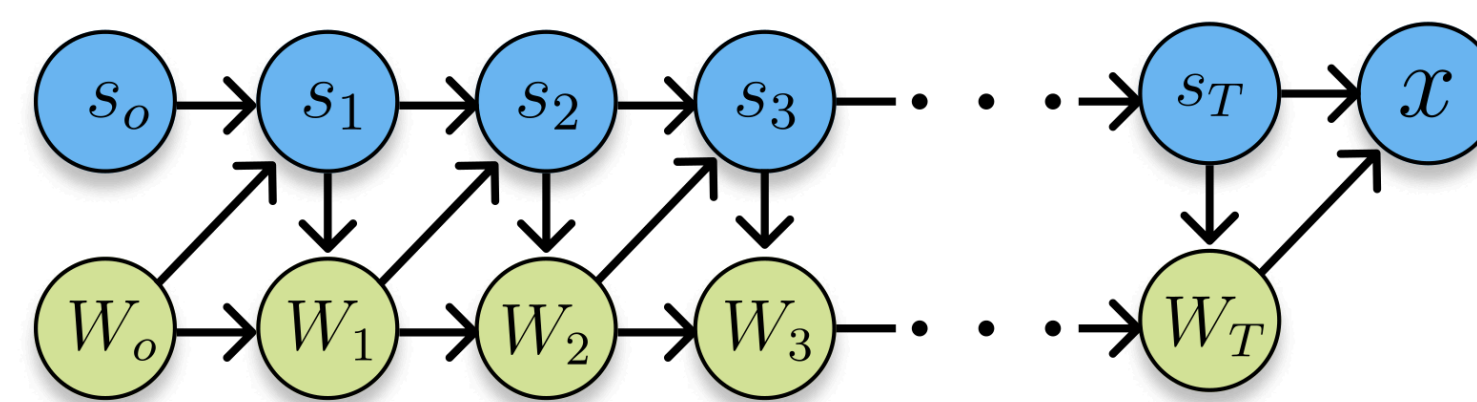


Figure 7: The **lifted** state graph: a memory row W_t is carried alongside each state. The **state-only** process may be non-Markovian, yet the **joint** (s_t, W_t) stays Markovian.

We instantiate φ per task so that the memory **disambiguates** aliased states:

- **LINES**: $\Omega = \mathbb{R}^d$, $\varphi(s, a) = (w_a^\top \psi(s)) v_a$ —the linear path-dependent policy.
- **GRAPHS**: carry the information of **when** each node’s it last changed—exactly the signal a plain graph network misses.

A single architecture works across tasks: a **self-referential weight matrix (SRWM)**,

$$q_t, k_t, v_t, \beta_t = W \psi(s_t) + b, \quad \bar{v}_t = W_{t-1} \zeta(k_t), \quad (8)$$

$$W_t = W_{t-1} R + \sigma(\beta_t)(v_t - \bar{v}_t) \oplus \zeta(q_t), \quad (9)$$

with slow weights W, b (SGD) and fast memory W_{t-1} at self-set rate $\sigma(\beta_t)$. The rotation R diversifies W_t ’s eigenvectors, **reducing aliasing and improving exploration**; it beats similarly-sized LSTMs/GRUs.

Training objectives still apply. Because the latent transitions are **deterministic** and the lifted graph remains **finite**, the **same** losses that guarantee $p_F^T(x) \propto R(x)$ on the plain graph **still** guarantee it on the lifted graph (proved for all popular objectives, including TB).

V. EMPIRICAL RESULTS

We evaluate on grid exploration, the **lazy random walk**, set generation, and sequence design; across all tasks **path-dependent** yields a **more adequate fit** than its Markovian counterpart.

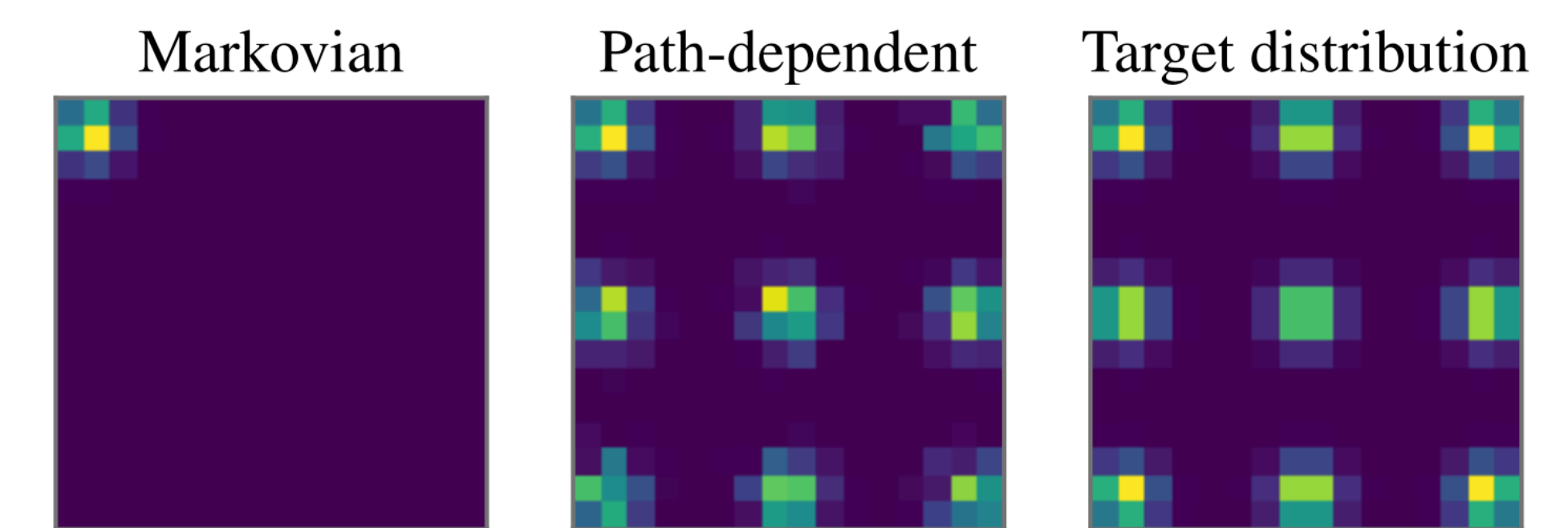


Figure 8: **Grid World**. **path-dependent** recovers the multi-modal target; **Markovian** collapses onto a few modes.

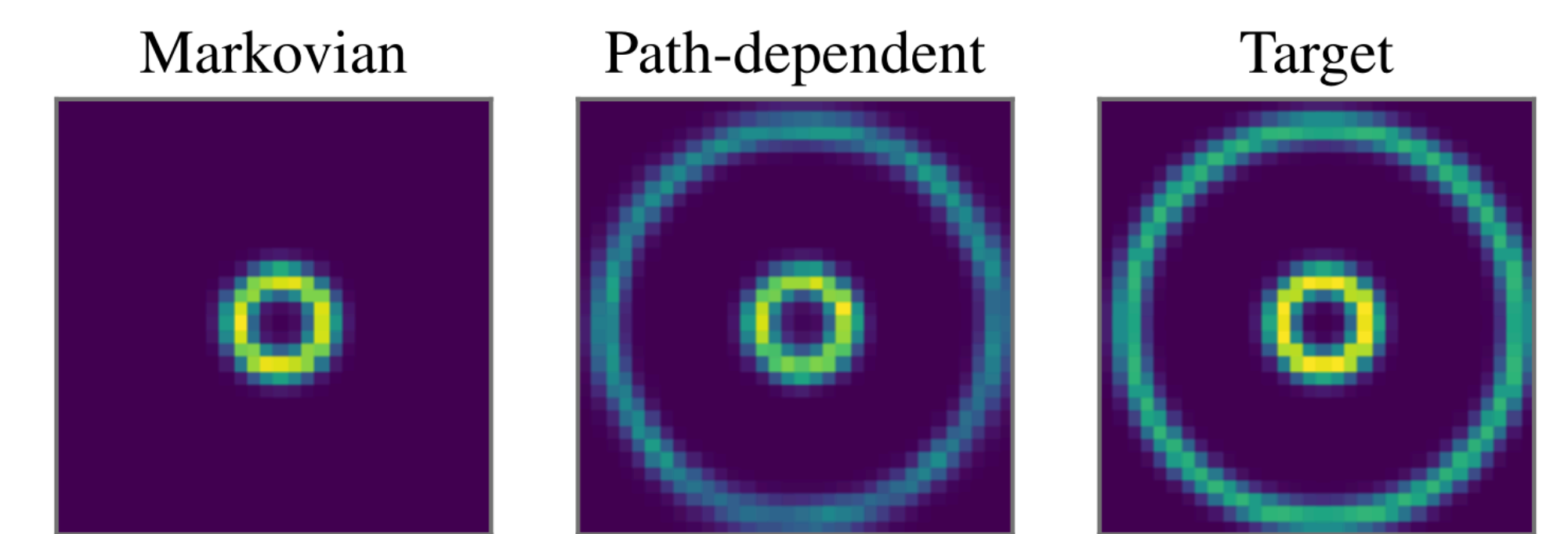


Figure 9: **Lazy random walk**. Learned densities: **Markovian** recovers only the inner ring, while **path-dependent** matches both rings of the target.

On the **lazy random walk** (appendix), the **recurrent memory** closes the gap: the self-referential weight matrix (R-SRWM) **beats similarly-sized LSTMs and GRUs**, and both its **rotation matrix** and **recurrent link** are essential.

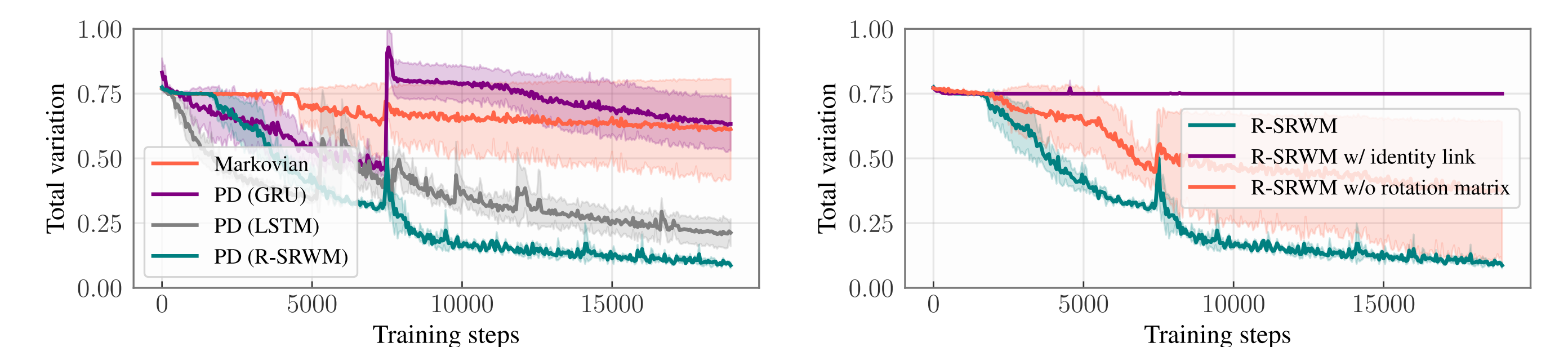


Figure 10: **R-SRWM** beats PD-GRU / PD-LSTM and Figure 11: **Ablations**: rotation and recurrence both matter.

VI. TAKE-AWAYS

- **Markovian samplers alias states**—provably and empirically **less expressive**.
- Root cause: the environment becomes (nearly) **partially observable**.
- We fix this by attaching a **learned recurrent memory** to the state, keeping the joint process Markovian so **standard training carries over unchanged**.
- Result: **faster, stabler convergence** and a **better fit** to the target.